```
DIM = 4

CONV_SECTION
(  1)  1  4  9  6
(  2)  4  1 10  6
(  3)  9  4  1 10
(  4) 10  1  4  9
(  5)  1 10  1  9

END

strong validity table :
\ I       |       |
 \ N      |       |
P \ E     |       |
 O \ Q    | 1     | #
  I \ S   |       |
   N \    |       |
    T \   |       |
     S \  |       |
---------------------
1         | *..*. :  2
2         | .*.*. :  2
3         | ..*.* :  2
4         | .*..* :  2
5         | *.*.. :  2
            ...........
#         | 22222
```

So the polytope is in fact a pentagon, as K_2 should be....

9.3 Constructing the Permuto-Associahedron

We will now describe the construction of the permuto-associahedron of Kapranov [313], as recently achieved in [453] (Example 0.10). There are analogous objects constructed for signed, bracketed permutations in [453], but we do not discuss those here. Also our discussion skips some details: we refer to [453] for the "missing pieces."

The construction depends on a version of the associahedron in especially nice coordinates. For this, we define

$$\Delta_n^f \ := \ \mathrm{conv}\{\boldsymbol{f}_0, \boldsymbol{f}_1, \ldots, \boldsymbol{f}_n\}, \qquad \text{for } \boldsymbol{f}_i := \boldsymbol{e}_1 + \ldots + \boldsymbol{e}_i,\ \boldsymbol{f}_0 = \mathbf{0},$$

as a reference simplex. Again we use the special "cyclic" $(n+1)$-gon

$$C_2(n+1) = \mathrm{conv}\{\binom{i}{i^2} : 0 \le i \le n\}.$$

Proposition 9.12. *Consider the linear projection map*

$$\pi : \quad \Delta_n^f \longrightarrow C_2(n{+}1), \qquad \boldsymbol{f}_i \longmapsto \binom{i}{i^2},$$

which maps $\boldsymbol{e}_i \longmapsto \binom{1}{2i-1}$ *for* $i \geq 1$. *Its scaled fiber polytope*

$$K_{n-2}^f \quad := \quad 3\binom{n+1}{3} \cdot \Sigma(\Delta_n^f, C_2(n+1))$$

has integral vertices, given by

$$\boldsymbol{v}^T \;:=\; \sum_{[i,j,k]\,\in\, T} \tfrac{1}{2}(j-i)(k-i)(k-j)\cdot(\boldsymbol{f}_i + \boldsymbol{f}_j + \boldsymbol{f}_k) \;\in\; \mathbb{Z}^n,$$

for all triangulations T *of* $C_2(n+1)$ *without new vertices. Here the sum is over all triples* $i < j < k$ *such that* $\big(\pi(\boldsymbol{f}_i), \pi(\boldsymbol{f}_j), \pi(\boldsymbol{f}_k)\big)$ *is a triangle in the triangulation* T, *of area* $\frac{1}{2}(j-i)(k-i)(k-j)$.

Furthermore, all vertices lie on a sphere around the origin:

$$\sum_{i=1}^{n} (\boldsymbol{v}_i^T)^2 \;=\; \binom{n+1}{3}\frac{30n^4 - 33n^2 + 2}{70}.$$

A linear description of K_{n-2}^f *is given by the equations*

$$\begin{aligned}
\sum_{i=1}^{n} x_i \quad &= \quad 3\binom{n+1}{3}\frac{n}{2} &&= \frac{n^2(n^2-1)}{4}\\
\sum_{i=1}^{n} (2i-1)\cdot x_i \quad &= \quad 3\binom{n+1}{3}\frac{6n^2+1}{15} &&= \frac{6n^5 - 5n^3 - n}{30},
\end{aligned}$$

which describe the $(n{-}2)$*-subspace of* $\mathbb{R}^n$ *that contains* K_{n-2}^f, *and facet-defining inequalities*

$$\boldsymbol{c}^{ij}\boldsymbol{x} \;:=\; \sum_{k=i+1}^{j} \Big((-2k+1) \;+\; i \;+\; j\Big)\, x_k \quad \geq \quad \binom{j-i+1}{3}\frac{3(j-i)^2 - 2}{10}$$

for $0 \leq i < j \leq n$ *and* $2 \leq j - i \leq n-1$.

This is just what we worked out in Example 9.11, after the linear transformation in $\mathbb{R}^n$ that sends $\boldsymbol{e}_i \longmapsto \boldsymbol{f}_i$ for $0 \leq i \leq n$ (that is, $\boldsymbol{e}_0 = \boldsymbol{0}$ to $\boldsymbol{f}_0 = \boldsymbol{0}$).

The *magical* little thing is that the vertices lie on a sphere. One can prove this by analyzing the situation along an edge, corresponding to a single rebracketing/change-of-diagonal, but there is no really good (that is, geometric) explanation. Do you have any ideas? The strange thing is that the whole construction of fiber polytopes is certainly affinely invariant: but suddenly here we get an effect that is decidedly nonlinear, since affine transformations distort the unit sphere.

Now the associahedron, in the coordinates of Proposition 9.12, is used to construct the permuto-associahedron.

Definition 9.13. (Kapranov [313])
The *face lattice of the permuto-associahedron* $K\Pi_{n-1}$ is a partially ordered set, defined as follows.

The elements of $K\Pi_{n-1}$ are ordered partitions of $\{1, 2, \ldots, n\}$ into at least two parts, *partially bracketed*: this means that the blocks are treated as if they were being multiplied together, and some of them are grouped together by brackets to indicate order of multiplication. In particular, every pair of brackets encloses at least two blocks.

The order relation on these bracketed partitions is as follows: $A \leq B$ if and only if B is obtained from A by removing pairs of brackets and possibly combining all the blocks within it into one block (if there are no brackets inside the pair we are considering).

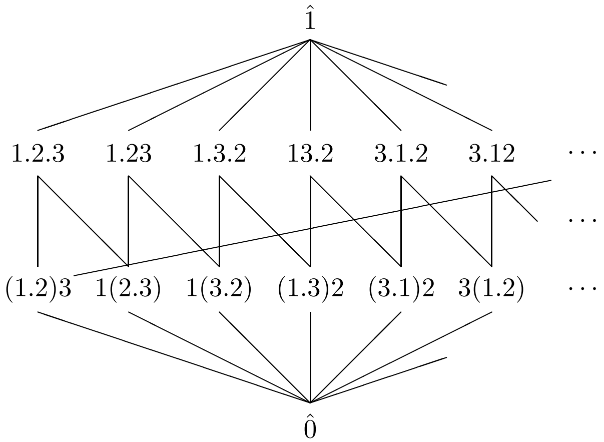
Finally, an extra minimal element $\hat{0}$ is included in $K\Pi_{n-1}$.

This yields a large, combinatorially defined poset. Typical elements (for $n = 7$) that are comparable in $K\Pi_6$ are

$$((4.3)((5.7)1))(6.2) \quad < \quad (3\,4.1\,5\,7)6.2 \quad < \quad 3\,4.1\,5\,7.6.2\,.$$

One can show quite easily that $K\Pi_{n-1}$ is in fact a graded, atomic, and coatomic lattice of length n. Thus it "looks like" the face lattice of an $(n-1)$-polytope. The coatoms ("facets") of $K\Pi_{n-1}$ are the ordered partitions of $\{1, \ldots, n\}$, without brackets. The atoms ("vertices") correspond to complete parenthesizations of permutations of the letters $1, 2, \ldots, n$. The edges are of two types: they correspond either to a single reparenthesization, or to a transposition of two adjacent letters that are grouped together. For $n = 3$ we get the face lattice of a 12-gon, as follows:

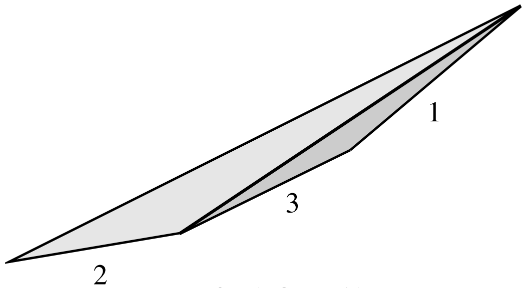
$K\Pi_2$:



Kapranov [313] used quite heavy machinery that to show that $K\Pi_{n-1}$ is the face poset of a "cellular ball"; a simpler argument is also in [453, Sect. 2]. In the following we want to show the stronger result that this is the face lattice of a convex $(n-1)$-polytope, which we will also denote by $K\Pi_{n-1}$.

Our large figure (on the next page) shows a drawing of $K\Pi_3$, as a polytope. A few vertices have been labeled by the corresponding completely bracketed permutations — you might continue this a little: for fun or to figure out how the combinatorial description matches the geometry. (However, don't scribble in the book if it is not yours!)

Example 9.14 (Permuto-associahedron). [453]
How do we get a vertex $\boldsymbol{v}^\alpha$ for every bracketed permutation α? For this we rewrite α as a pair $\alpha = (\sigma, T)$, where σ is a permutation of $\{1, \dots, n\}$, and $T = T(\alpha) \subseteq \binom{\{0,\dots,n\}}{3}$ is the set of triples of the triangulation of $C_2(n+1)$ that is given by the bracketing of α. With this we interpret the bracketed permutation as a triangulation of the $(n+1)$-gon $C_2(n+1)$, whose lower edges are labeled by $\sigma(1), \dots, \sigma(n)$ — for example, 2(3.1) is represented by



2 (3 . 1)

Now every permutation $\sigma = \sigma(1)\sigma(2)\dots\sigma(n)$ determines a simplex in $\mathbb{R}^n$, namely

$$\begin{aligned} \Delta_n(\sigma) \quad &:= \quad \mathrm{conv}\{\mathbf{0},\ \boldsymbol{e}_{\sigma(1)},\ \boldsymbol{e}_{\sigma(1)}+\boldsymbol{e}_{\sigma(2)},\ \ \dots\ ,\boldsymbol{e}_{\sigma(1)}+\dots+\boldsymbol{e}_{\sigma(n)} = \mathbf{1})\} \\ &= \quad \{\boldsymbol{x} \in \mathbb{R}^n : 1 \ \geq\ x_{\sigma(1)}\ \geq\ x_{\sigma(2)}\ \geq\ \dots\ \geq x_{\sigma(n)}\}. \end{aligned}$$

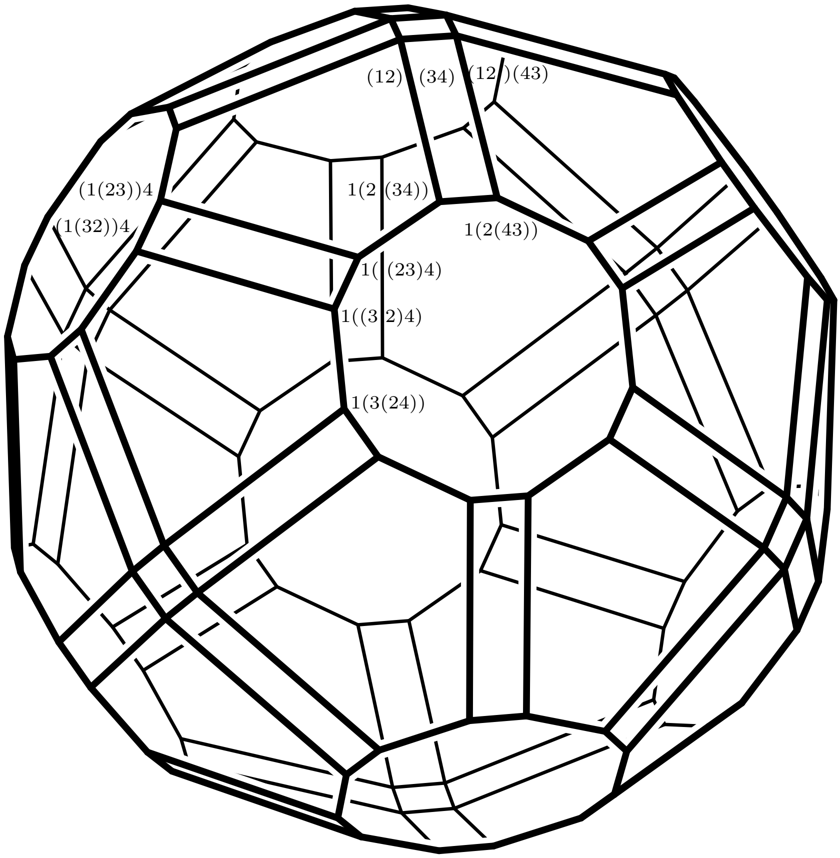
Thus $\Delta_n(\sigma)$ is just the convex hull of the section

$$\gamma^\sigma : \ [0,n] \quad \longrightarrow \quad \mathbb{R}^n$$

that we have associated to σ in Example 9.8. For example, the permutation $\sigma = 12\dots n$ determines the "standard simplex"

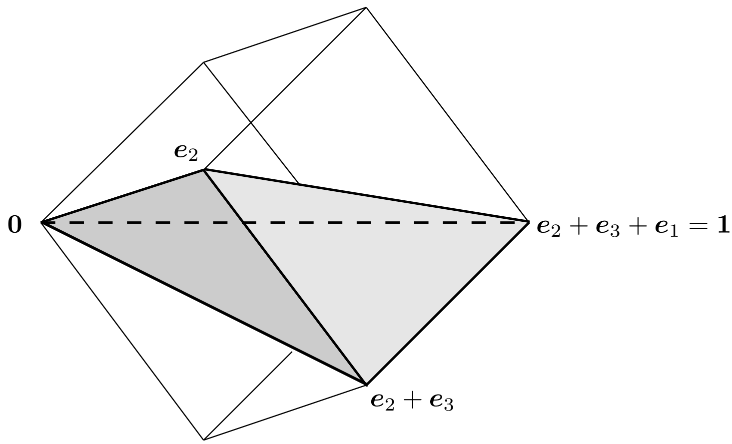
$$\Delta_n^f \quad = \quad \mathrm{conv}\{\boldsymbol{f}_1, \dots, \boldsymbol{f}_n\}.$$

The description of the simplices $\Delta_n(\sigma)$ in terms of their inequality systems also shows that they fit nicely together to form a triangulation of the unit cube $[0,1]^n$.



The permuto-associahedron $K\Pi_3$
(in wonderful postscript graphics by Jürgen Richter-Gebert,
generated from PORTA output).

As an example, the simplex corresponding to the permutation 231 for $n = 3$ is shown in the following figure:



2 3 1

Each of these simplices has a natural map down to $\mathbb{R}^2$, where we map

$$\begin{array}{rcll} \pi_\sigma : \quad \Delta_n(\sigma) & \longrightarrow & C_2(n+1), & \\ e_{\sigma(1)}+\ldots+e_{\sigma(i)} & \longmapsto & \begin{pmatrix} i \\ i^2 \end{pmatrix}, & \quad \text{for } 0 \le i \le n. \end{array}$$

Since the simplices fit together so nicely to form a triangulation of the cube $[0,1]^n$, and the projection maps are defined consistently on the vertices, we obtain a *continuous, but nonlinear* "folding map"

$$\Pi : \ [0,1]^n \ \longrightarrow \ C_2(n+1),$$

which is linear on the simplices $\Delta(\sigma)$. Furthermore, for every bracketed permutation, there is an obvious section to this folding map! For this we define the section on the vertices by

$$v_i \ = \ \begin{pmatrix} i \\ i^2 \end{pmatrix} \ \longmapsto \ e_{\sigma(1)}+\ldots+e_{\sigma(i)}$$

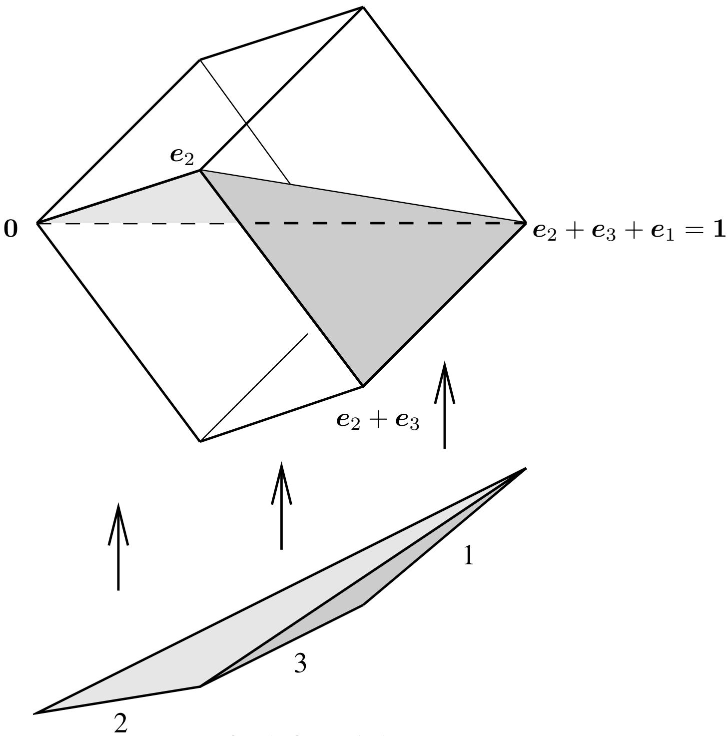
and then extend linearly on the triangles of the triangulation of $C_2(n+1)$, to get a section

$$\gamma^\alpha : \ C_2(n+1) \ \longrightarrow \ [0,1]^n$$

associated with the string $\alpha = (\sigma, T)$.

The integral over this section defines a point in $\mathbb{R}^n$ for the completely bracketed permutation α, and this creates the vertices of the permuto-associahedron.

Our figure illustrates the section $\gamma^\alpha : C_2(4) \longrightarrow [0,1]^3$ that is associated to the bracketed permutation $\alpha = 2(3.1)$ by this method.



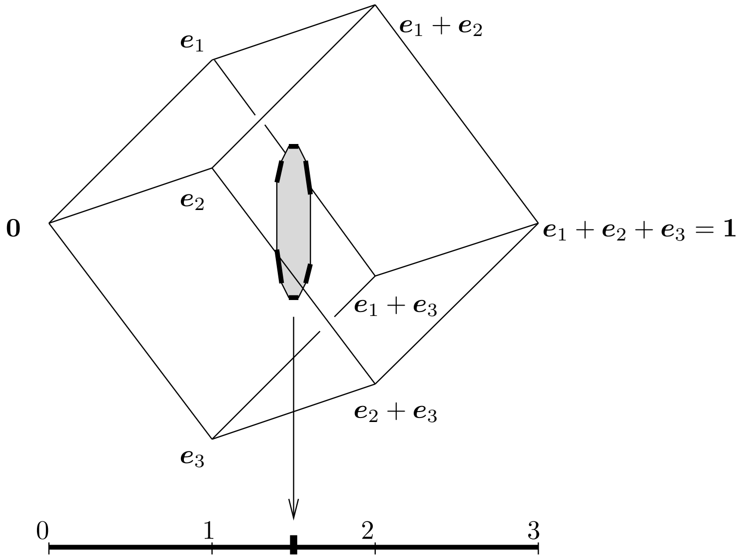
2 (3 . 1)

Another way to view this construction is the following. The vertices of the "special associahedron" K^f_{n-2} in Proposition 9.12 satisfy $v_1 > v_2 > \ldots > v_n$: in fact, the *little associahedron*

$$\frac{1}{3\binom{n+1}{3}} K^f_{n-2} \quad = \quad \Sigma(\Delta^f_n, C_2(n+1))$$

is a fiber polytope, and thus it lies in the corresponding simplex Δ^f_n of the triangulation of the cube $[0,1]$ we considered. Now by just permuting coordinates, we get $n!$ copies of the little associahedron in the various simplices, and the convex hull of those $n!$ little associahedra is the permuto-associahedron.

Our figure tries to sketch this for $n = 3$, where the associahedra are six little line segments, whose convex hull is a 12-gon. Again the drawing is not metrically correct — it is a mere sketch of the geometric situation, trying to illustrate the position of the "little associahedra" within the n-cube, and how their convex hull forms the $(n-1)$-dimensional associahedron. ("The idea is the important thing," as Mr. Lehrer would say.) In contrast to this, our big picture on page 314 of the 3-dimensional permuto-associahedron was computer generated from the actual coordinates we gave for its position in $\mathbb{R}^4$ — thus it represents the actual geometry of the polytope, not only its combinatorics.



The only change we do for the formulas is that instead of the average integral we take three times the integral, that is, we blow the polytope up by a factor $3\binom{n+1}{3} = 3\mathrm{vol}(C_2(n+1))$, in order to get integral coordinates.

Theorem 9.15. *The formula*

$$\begin{aligned} \boldsymbol{v}^\alpha &:= 3\int_{C_2(n+1)} \gamma^\alpha \mathrm{d}x\,\mathrm{d}y \\ &= \sum_{(i,j,k)\in T(\alpha)} \frac{1}{2}(j-i)(k-i)(k-j)\cdot(\boldsymbol{f}_{\sigma(i)} + \boldsymbol{f}_{\sigma(j)} + \boldsymbol{f}_{\sigma(k)}) \end{aligned}$$

associates a point $\boldsymbol{v}^\alpha \in \mathbb{Z}^n$ *to every completely bracketed permutation* α. *The polytope*

$$\mathrm{conv}\,\{\boldsymbol{v}^\alpha : \alpha = (\sigma, T(\alpha)) \textit{ a completely bracketed permutation of } [n]\}$$

is the permuto-associahedron, that is, its face lattice is isomorphic to the poset $K\Pi_{n-1}$ *of Definition 9.13, under the correspondence* $\alpha \longmapsto \boldsymbol{v}^\alpha$. *It is* $(n-1)$*-dimensional, contained in the hyperplane*

$$\Big\{\boldsymbol{x} \in \mathbb{R}^n : \sum_i x_i = \frac{n^4 - n^2}{4}\Big\}.$$

Furthermore the vertices of the permuto-associahedron are integral in this coordinatization, and they all lie on the sphere around the origin:

$$\sum_{i=1}^{n} (\boldsymbol{v}_i^\alpha)^2 \;=\; \binom{n+1}{3} \frac{30n^4 - 33n^2 + 2}{70}.$$

Proof. A proof with details is in [453], and we refer to the treatment there. What is the idea? First the associahedron in Proposition 9.12 lies in the hyperplane $H = \{\boldsymbol{x} \in \mathbb{R}^n : \mathbb{1}\,\boldsymbol{x} = \frac{n^4-n^2}{4}\}$. Since the vertices of $K\Pi_{n-1}$ can be generated by permuting the coordinates of vertices of K_{n-2}^f, we get that our polytope $K\Pi_{n-1}$ is also contained in H.

Then we obtain the facet-defining inequalities. That is, to every ordered partition ϕ of $[n]$ we associate a linear function $\boldsymbol{c}^\phi$. For this let ϕ have p blocks, and write it as

$$\phi \;=\; \sigma(i_1)\cdots\sigma(j_1)\;.\;\sigma(i_2)\cdots\sigma(j_2)\;.\;\cdots\;\cdots\;.\;\sigma(i_p)\cdots\sigma(j_p),$$

where the numbers i_r and j_r just tell us about the "block structure" of ϕ, with

$$1 = i_1 \le j_1, \quad j_1 + 1 = i_2 \le j_2, \qquad \ldots \qquad , j_{p-1} + 1 = i_p \le j_p = n.$$

Then the linear function $\boldsymbol{c}^\phi$ we need is given by

$$c_k^\phi \;=\; i_r + j_r \qquad \text{if } i_r \le \sigma^{-1}(k) \le j_r,$$

that is, if the letter k lies in the rth block of ϕ.

Now it is not too hard to show (if you use the explicit description of the associahedra in Proposition 9.12, and the symmetry of the situation) that the function $\boldsymbol{c}^\phi$ is minimized exactly by those vertices $\boldsymbol{v}^\alpha$ with $\alpha \le \phi$ (in the lattice $K\Pi_{n-1}$), that is, it defines a facet with exactly the right vertices on it.

Now we have to argue that we have found all the facet-defining inequalities. One way to do this is to use a lot of combinatorics of the poset $K\Pi_{n-1}$, such as that every element of rank $n-1$ lies on exactly two coatoms (facets), together with Exercise 2.8(iv). Such an argument — for the associahedron — is in [266]. The alternative is to use that the convex hull of the vertices that we have constructed is contained in a larger polytope given by the inequalities that we have found. Now one can show that for every linear function, one of "our" vertices maximizes the linear function over the facet-defining inequalities we have found. This shows that our description of $K\Pi_{n-1}$ by inequalities is complete.

The final fact one then needs is that the vertex-facet incidences already determine the polytope — see Exercise 2.7. Thus, we have constructed a polytope with the "right" face lattice $K\Pi_{n-1}$. □

9.4 Toward a Category of Polytopes ?

Fiber polytopes form an important first step in investigating a "category of polytopes," a program suggested by Louis Billera. In fact, this category should have interesting properties that are fundamental to many geometric questions. It is surprising that the basic "universal constructions" for such a category have hardly been studied. Among them are the *fiber polytopes*, which form "kernel objects"; the *mapping polytopes* discussed below, which are "spaces of maps"; and *cofiber polytopes* that should play the role of "co-kernel objects" — for which we lack even a good definition (Problem 9.16*).

Definition 9.16. Let $P \subseteq \mathbb{R}^p$ and $Q \subseteq \mathbb{R}^q$ be full-dimensional polytopes (of dimensions p and q). Then the set of affine maps $f^{A,\boldsymbol{z}} : \mathbb{R}^p \longrightarrow \mathbb{R}^q$, $\boldsymbol{x} \longmapsto A\boldsymbol{x} + \boldsymbol{z}$ can be identified with $\mathbb{R}^{q\times p} \times \mathbb{R}^q \cong \mathbb{R}^{(p+1)q}$. The subset

$$\Phi(P,Q) \quad := \quad \{(A,\boldsymbol{z}) \in \mathbb{R}^{(p+1)q} : f^{A,\boldsymbol{z}}(P) \subseteq Q\}$$

is a polytope in $\mathbb{R}^{(p+1)q}$ of dimension $(p+1)q$: the *mapping polytope* of the pair (P,Q).

We omit the (easy) proof that $\Phi(P,Q)$ actually is a full-dimensional polytope (see [450]). Also there is a natural generalization to polyhedra. Here we note a few important examples.

Examples 9.17.

(i) When P is a point ($p = 0$), then $\Phi(P,Q)$ is isomorphic to Q. More generally, if $P = \Delta_p$ is a simplex with $p+1$ vertices, then the affine image of the vertices of P can be chosen independently in Q, which proves an affine equivalence

$$\Phi(\Delta_p, Q) \quad \cong \quad Q^{p+1}.$$

(ii) In particular, take the $(d-1)$-simplex $\Delta_{d-1} = \mathrm{conv}\{\boldsymbol{e}_1,\ldots,\boldsymbol{e}_d\} \subseteq \mathbb{R}^d$ on the hyperplane $H := \{\boldsymbol{x} \in \mathbb{R}^d : \mathbb{1}\,\boldsymbol{x} = 1\}$, as before. Then we can identify affine maps with $H \longrightarrow H$ with linear maps $\mathbb{R}^d \longrightarrow \mathbb{R}^d$ that fix $\mathbf{0}$. Such maps are given by a matrix $V = (\boldsymbol{v}_1,\ldots,\boldsymbol{v}_d) \in \mathbb{R}^{d\times d}$, where $\boldsymbol{e}_i \longmapsto \boldsymbol{v}_i$. Thus the mapping polytope $\Phi(\Delta_{d-1},\Delta_{d-1})$ turns out to be

$$\begin{aligned}
\Phi(\Delta_{d-1},\Delta_{d-1}) \quad &= \quad \{(\boldsymbol{v}_1,\ldots,\boldsymbol{v}_d) \in \mathbb{R}^{d\times d} : \ \boldsymbol{v}_i \in \Delta_{d-1} \text{ for } 1 \le i \le d\} \\
&= \quad \{(v_{ij}) \in \mathbb{R}^{d\times d} : \ v_{i,j} \ge 0 \text{ for } 1 \le i,j \le d, \\
&\qquad\qquad\qquad\qquad \sum_j v_{ij} = 1 \text{ for } 1 \le i \le d\} \\
&= \quad (\Delta_{d-1})^d.
\end{aligned}$$